



Bachelor Thesis

Quantum network simulator for quantum key distribution

by

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June 3, 2026

*Submitted in partial fulfillment of the requirements for
the VU degree of Bachelor of Science*

Quantum network simulator for quantum key distribution

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ABSTRACT

Context. Write this at the end

Goal. Write this at the end

Method. Write this at the end

Results. Write this at the end

Conclusions. Write this at the end

1 INTRODUCTION

The secure transmission of sensitive data is one of the key challenges of the modern digital era. Classical cryptographic algorithms such as RSA [6] face a significant threat from the emergence of quantum computers, as their security relies on the assumption that no efficient algorithm exists to invert the encryption using publicly available information. Quantum Key Distribution (QKD) has emerged as a provably secure alternative, grounded in the laws of quantum physics rather than computational complexity.

Many QKD protocols exist [1, 2, 9] that enable the generation of secure keys by exploiting the principles of quantum mechanics. This thesis focuses on implementing the E91 protocol due to its use of entangled qubits and Bell-state verification.

Transitioning these theoretical protocols into scalable real-world networks presents significant engineering challenges. For this reason, quantum network simulators such as SeQUeNCe [12] and QKDNetSim [5] are valuable tools, as they enable controlled testing of new protocols. Nevertheless, the development of such simulators introduces additional challenges, including the accurate modeling of entanglement and distance-dependent fidelity degradation. Due to these limitations and the lack of consensus among existing models, several assumptions were made, which are discussed later in this thesis.

This thesis addresses the following research question: *How can we design and implement an event-driven quantum network simulator capable of evaluating the performance, error-correction overhead, and adversarial resilience of the E91 protocol under realistic hardware imperfections?*

To answer this question, we present the design and implementation of a modular, high-performance quantum network simulator written in Rust [3]. Rust was chosen for its memory safety, high performance, and concurrency support without the overhead of a garbage collector. The core architecture is based on a discrete event-driven simulation engine that uses a priority queue to process events at picosecond resolution (ps). The platform models realistic quantum channels, including distance-dependent entanglement degradation, optical fiber attenuation, and detector cooldown states. It also implements a complete classical post-processing pipeline, including basis reconciliation, CHSH inequality testing, and Cascade error correction. Furthermore, a reactive web interface built

with Leptos and compiled to WebAssembly provides interactive real-time visualization of network topologies.

Through targeted simulation scenarios, we evaluate the performance of the E91 protocol over distances of up to approximately 150 km, assuming a 0.14% fidelity degradation per km based on [7]. To validate the security subsystem, we introduce an active eavesdropper model ("Mallory") performing interception attacks on transmitted qubits. The simulator demonstrates that such interference destroys quantum correlations, reducing the CHSH parameter S below the classical bound ($S < 2$) and triggering protocol termination.

The main contributions of this thesis are as follows:

- ★ **Event-Driven Simulation Engine:** We design and implement a high-performance discrete-event simulation engine in Rust using a priority queue. The engine resolves events at picosecond resolution, accurately modeling asynchronous node behavior, qubit propagation delays, and detector cooldown dynamics.

- ★ **Realistic Quantum Channel and Device Modeling:** We implement detailed abstractions of optical channels and quantum devices, including transmission probability $P_a = e^{-\alpha L}$, distance-dependent fidelity degradation, detector cooldown, and Poisson-distributed dark counts.

- ★ **End-to-End E91 Post-Processing and Security Verification:** We construct a complete pipeline for key sifting, CHSH inequality testing, and multi-round Cascade error correction. The simulator reliably detects eavesdropping attempts through degradation of the Bell parameter ($S < 2$).

The remainder of this thesis is organized as follows: Section 2 introduces quantum entanglement, the E91 protocol, and the Cascade error correction algorithm. Section 3 describes the simulator architecture and development challenges. Section 4 presents simulation results and their relevance to real-world scenarios. Finally, Section 7 compares the proposed simulator with existing tools before the concluding section.

2 BACKGROUND

Before discussing the design and evaluation of the simulator, this section introduces the fundamental quantum concepts underlying the E91 key distribution protocol.

In E91, as in any quantum communication system, information is transmitted using **quantum states**. These states encode probability amplitudes whose squared magnitudes determine the likelihood of measurement outcomes. A quantum state that is expressed as a linear combination of basis states is said to be in a **superposition**.

The simplest representation of a quantum system is the **qubit**, a two-level quantum state defined in the computational basis as:

$$|0\rangle, \quad |1\rangle.$$

Using Born's rule, the state of a qubit can be expressed as a linear combination of these basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where $\alpha, \beta \in \mathbb{C}$ and satisfy the probability condition $|\alpha|^2 + |\beta|^2 = 1$.

To operate on qubits, we can use a set of quantum gates analogous to logic gates in classical computing. One example is the **Hadamard (H)** gate, which creates superposition states and is represented by the following matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

For multi-qubit systems, one of the most important gates is the **CNOT** (controlled-NOT) gate, which acts on a two-qubit system. Its matrix representation is given by:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Quantum states can also be represented as vectors in a complex Hilbert space. In this representation, the application of a quantum gate corresponds to matrix multiplication between the gate operator and the state vector.

In order to transform a qubit into a superposition state, where both measurement outcomes have equal probability, the Hadamard gate is applied to a qubit initially prepared in a computational basis state, such as $|0\rangle$.

Representing the basis state $|0\rangle$ as a vector:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

we apply the Hadamard gate via matrix multiplication:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

Entanglement, a key aspect of the E91 protocol, occurs when two qubits are prepared in a quantum state such that their measurement outcomes exhibit correlations, regardless of distance. These entangled two-qubit states are known as Bell states, named after physicist John Bell, who developed Bell's theorem to test the non-classical nature of such correlations.

One of the four maximally entangled two-qubit Bell states is defined as:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

In order to produce a Bell state, you first apply a Hadamard gate, followed by a CNOT gate.

When both qubits of the Bell state are measured in the same basis, the outcomes are perfectly correlated, giving identical results (either 00 or 11) despite each result being individually random. This basis-dependent correlation is the key resource exploited in the E91 protocol, enabling both key generation and eavesdropping detection through later comparison of measurement outcomes.

Building on these quantum mechanical principles, the E91 protocol enables two nodes in the quantum internet, to establish a shared secret key using entangled qubit pairs distributed by a quantum source.

E91 relies on the distribution of entangled states, typically Bell states, such that correlations between measurement outcomes can be used both for key generation and for detecting the presence of an eavesdropper.

In the standard setting, a trusted, usually referred as epr node, distributes entangled qubit pairs to both of the client nodes over a quantum channel. Each party independently measures their qubits using randomly chosen measurement bases. The choice of measurement basis plays a central role in both the generation of the key and the security analysis of the protocol.

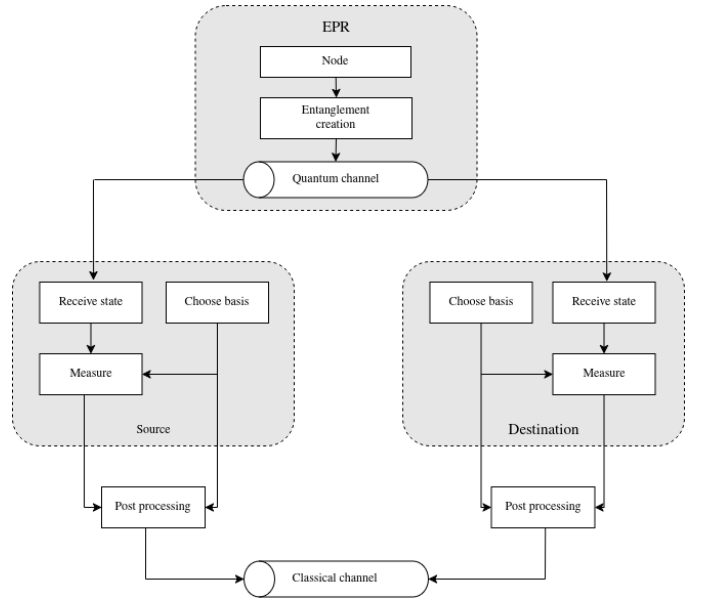


Figure 1: Diagram describing E91 process

Once all entangled pairs have been distributed and measured, both nodes share their chosen measurement bases over a classical channel, which reveals little to no useful information to an eavesdropper.

Using this basis information, the measured qubits can be divided into two groups:

- **Key qubits:** These correspond to measurements where both parties used the same basis. Assuming ideal entanglement and no fidelity loss, the resulting measurement outcomes should be perfectly correlated.
- **Test qubits:** These correspond to measurements performed in different bases. These outcomes are used to evaluate quantum correlations and to detect the presence of an eavesdropper in the communication channel.

In the E91 protocol, the measurement bases correspond to rotations on the Bloch sphere that define the detector settings used to measure incoming qubits. Following [4], the choice of measurement angles that maximizes quantum correlations and leads to a strong violation of the Bell inequality must be chosen in steps of $\frac{\pi}{8}$. The

ones chosen for this simulator were:

$$\begin{aligned}\alpha &= 0 \quad (0^\circ), & \alpha' &= \frac{\pi}{4} \quad (45^\circ), \\ \beta &= -\frac{\pi}{8} \quad (-22.5^\circ), & \beta' &= \frac{\pi}{8} \quad (22.5^\circ),\end{aligned}$$

where α, α' correspond to one party's polarizer settings and β, β' correspond to the other party's polarizer settings.

The CHSH test is an experimental test of Bell's theorem, which shows that no physical theory based on local hidden variables can reproduce all predictions of quantum mechanics.

This result is expressed using the CHSH S -parameter [8], defined as

$$S = |E(A_0, B_0) + E(A_0, B_1)| + |E(A_1, B_0) - E(A_1, B_1)|.$$

Here, $E(A_i, B_j)$ denotes the expectation value of the correlation between measurement settings A_i and B_j .

In any local hidden variable theory, the CHSH inequality imposes the bound

$$S \leq 2,$$

whereas quantum mechanics predicts a stronger limit,

$$S \leq 2\sqrt{2}.$$

Based on [10] when the basis are used as polarizer angles (moving the detector by such an angle) the expectation value can be calculated as:

$$E(a, b) = \cos(2(a - b))$$

Therefore, the CHSH value for our choice of basis is given by

$$\begin{aligned}E(\alpha, \beta) &= \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}, & E(\alpha, \beta') &= \cos\left(-\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}, \\ E(\alpha', \beta) &= \cos\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}, & E(\alpha', \beta') &= \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}.\end{aligned}$$

Substituting into the CHSH expression,

$$S = |E(\alpha, \beta) + E(\alpha, \beta')| + |E(\alpha', \beta) - E(\alpha', \beta')|,$$

gives

$$S = |\sqrt{2}| + |-\sqrt{2}| = 2\sqrt{2}.$$

If after calculating the S parameter, we get $|S| < 2$, we can assume that the correlation is not maximal and there is a third person in the communication

The Cascade protocol functions as a multi-pass block-parity check designed to reconcile bit errors between Alice and Bob caused by fidelity loss. In the first pass, the sifted bit strings are divided into equal-sized blocks (k_1), and Alice transmits the parity of each block to Bob. For any block where parities disagree—signaling an odd number of errors—the parties execute the BINARY subprotocol. This subprotocol bisects the block and compares the parities of the halves until the exact position of a single erroneous bit is isolated and corrected by Bob, leaving all remaining blocks with an even number of errors.

In subsequent passes ($i > 1$), the block size increases (k_i) and a new mapping function shuffles the bits into different groups. Alice and Bob exchange parities for these new blocks to identify remaining discrepancies. Correcting an error in a later pass alters the parity of the bits in earlier passes, triggering a feedback loop; the protocol identifies the smallest blocks from previous passes that

now have odd errors and recursively runs BINARY to correct them. For this simulator, this framework will run for exactly four rounds of cascade to ensure identical keys before proceeding to hash-based verification.

3 DESIGN

In this section, you would provide a high-level description of the system or solution and explain your design choices.

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4 EVALUATION

Discuss the design of your experiments, the results you obtained, and how they help in evaluating the claims you made in the introduction. You may also use the evaluation results in this section to justify your design choices or assess the contributions of different aspects of your design towards the overall goals.

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5 DISCUSSION

Here you put your results in context (possibly grouped by research question). Usually, this section focuses on analyzing the implications of the proposed work for current and future research and for practitioners.

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6 THREATS TO VALIDITY

Report about each type of threat to the validity of the experiment, according to the classification framework proposed by Wohlin *et al.* [11].

6.1 Internal Validity

6.2 External Validity

6.3 Construct Validity

6.4 Conclusion Validity

7 RELATED WORK

Describe here scientific papers similar to your experiment, both in terms of goal and methodology. One paragraph for each paper (we expect about 5-8 papers to be discussed). Each paragraph contains: (i) a brief description of the related paper and (ii) a black-on-white description about how your work differs from the related paper. You may place this section immediately after the Background section, if necessary.

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8 CONCLUSION

Briefly summarize your contributions, and share a glimpse of the implications of this work for future research.

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